



聚苯硫醚材料物性表

聚苯硫醚产品系列

玻纤增强

牌号：WG402S-35S-SJ

中国.重庆



重庆沃尔夫化工有限公司 PPS 材料物性表

Technical Data Sheet

牌号 (Grade) : PPS-WG402S-35S-SJ

聚苯硫醚 Polyphenylene sulfide

PPS-WG402S-35S-SJ 是采用 30%玻璃纤维增强改性制得的复合材料，具有高韧性、耐高温特点；本色料具有耐黄变的性能；同时具有良好的金属、环氧、硅胶、聚氨酯结合力，同时满足激光打码要求。

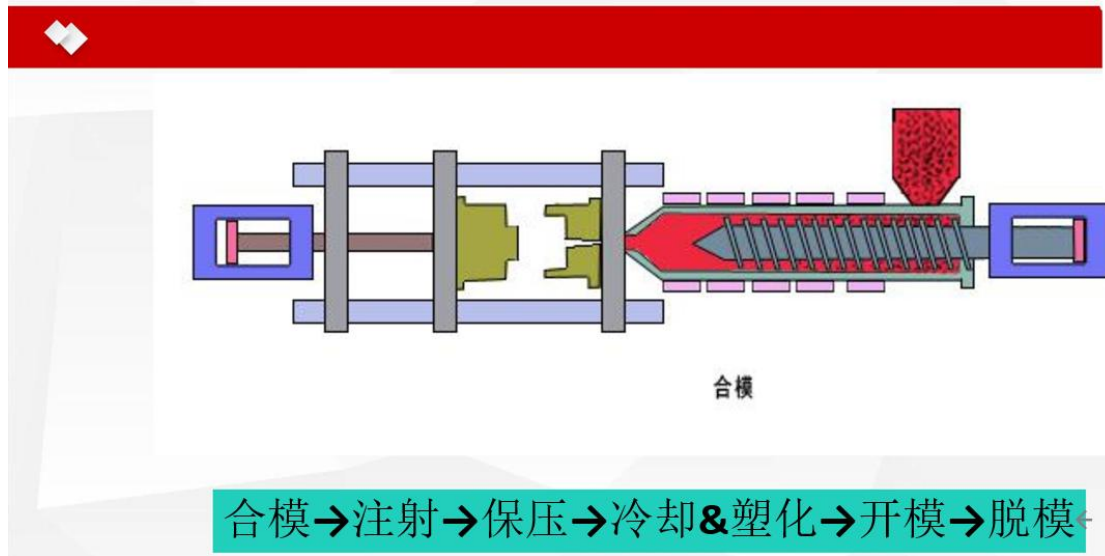
PPS-WG402S-35S-SJ is a composite material made by 30% glass fiber reinforcement modification, which has high toughness and high temperature resistance characteristics; The natural color material has the property of resistance to yellowing; Simultaneously possessing good adhesion between metal, epoxy, silicone, and polyurethane, while meeting the requirements of laser coding.

物理性能 (Physical Property)	检测标准(Test method)	单位 (Unit)	典型值(Typioal Value Unit)
比重 (Specific Gravity)	ISO 1183	g/cm ³	1.54
成型收缩率 (Molding Shrinkage)			
平行方向 (Flow)	GB/T 15585	%	0.32
垂直方向 (Across Flow)	GB/T 15585	%	0.67
吸水率 (Water Absorption) 23℃, 24h	ISO 62	%	0.02
机械性能 (Mechanical Property)	检测标准(Test method)	单位 (Unit)	典型值(Typioal Value Unit)
拉伸强度 (Tensile Strength)	ISO 527	Mpa	155
断裂伸长率 (Tensile Elongation)	ISO 527	%	2.2
弯曲模量 (Flexural Modulus)	ISO 178	Gpa	9.5
弯曲强度 (Flexural Strength)	ISO 178	Mpa	230
缺口冲击强度 (Notched Izod Impact)	ISO 179	KJ/m ²	10
热性能 (Thermal Property)	检测标准(Test method)	单位 (Unit)	典型值(Typioal Value Unit)
熔点 (Melting Point)	ISO 11357	℃	282
热变形温度 (Thermal deformation temperature)	ISO 75	℃	260
RTI (relative temperature index)	ASTM	℃	180
阻燃性 (Flame retardancy)	UL-94	class	V-0
电性能 (Electrical Properties)	检测标准(Test method)	单位 (Unit)	典型值(Typioal Value Unit)
介电强度 (Dielectric Strength)	IEC 60243	KV/mm	20
介电常数 (Dielectric Constant)	IEC 60250		4
损耗系数 (Loss Coefficient)	IEC 60250		0.003
体积电阻率 (Volume Resistivity)	IEC 60093	Ω . cm	2.2×10 ¹⁵
CTI (Comparative Tracking Index)	IEC 60112	V	175

The information in this data table is the typical analysis and testing values of Chongqing Wolf Chemical Co., Ltd. laboratory, and is not used as a basis for customer and others' acceptance. The specific standards depend on the customer to determine separately based on their own needs.



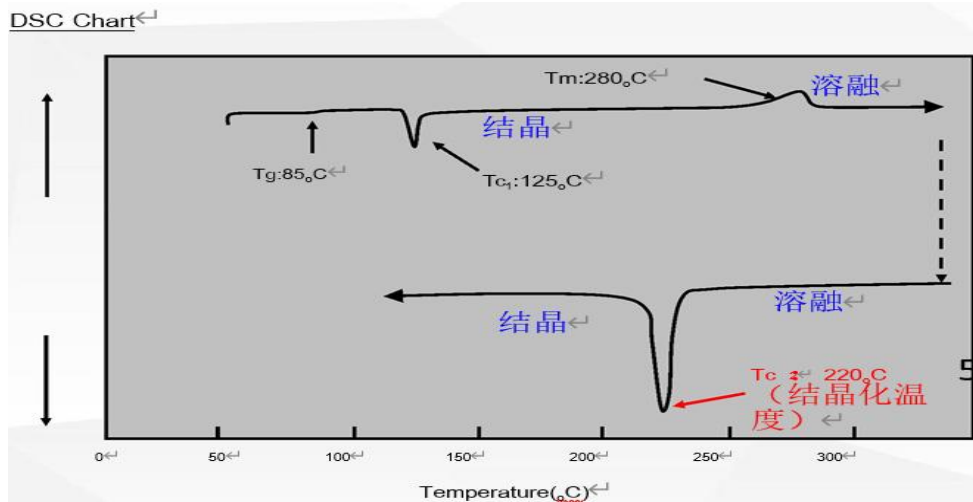
注塑流程



成型条件

树脂温度: 300~320℃ (熔点+20~30℃)
模具温度: 130~150℃ (PPS结晶速度慢, 低温影响结晶度)
干燥: 120℃, 3h (除去水分和挥发性组分)
填充速度: 30~100mm/sec
保压压力: 40~80MPa
保压时间: 浇口固化
冷却时间: 计量时间 + α
螺杆转速: 80~150rpm
螺杆背压: 2~5MPa
射出终点: 5~10mm

低模温的影响



从DSC可以看出PPS结晶温度高达220℃，模具温度过低，或导致结晶度不足，进而引发以下问题：

- (1) 强度不足；
- (2) 外观不良
- (3) 后收缩
- (4) 翘曲变形
- (5) 空洞、缩痕

更多产品信息及技术资料，请关注微信公众号“重庆沃尔夫化工有限公司”

